Friday, April 3, 2026 7:50 PM

* Linear Algebra → Halfspaces & Distance *

⇒ Recap
⇒ Shifting in 2D Lines
⇒ Weight vector Lavity to line & Hyperplane
⇒ Distance B/w origin & Line
⇒ Distance B/w any point & line
⇒ Halfspaces
=

+ve

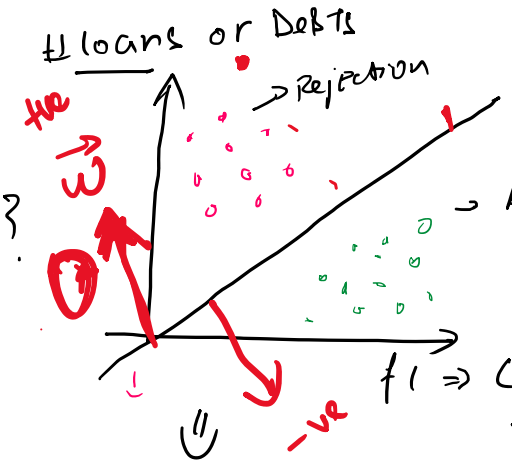
new poi

+v

(+) MOTIVATION ⇒

(*) Loan acceptance / Rejection?
=

(*) line, Boundary, Classifier
=



=

inference?

(+) → more loans → Rejection ↑
=
(*) → credit score ⇒ Acceptance ↑
=

eq^n of this line ⇒ $w_1 x_1 + w_2 x_2 + w_0 = 0$

(*)

**

$\Rightarrow w^T x + w_0 = 0 \Rightarrow$

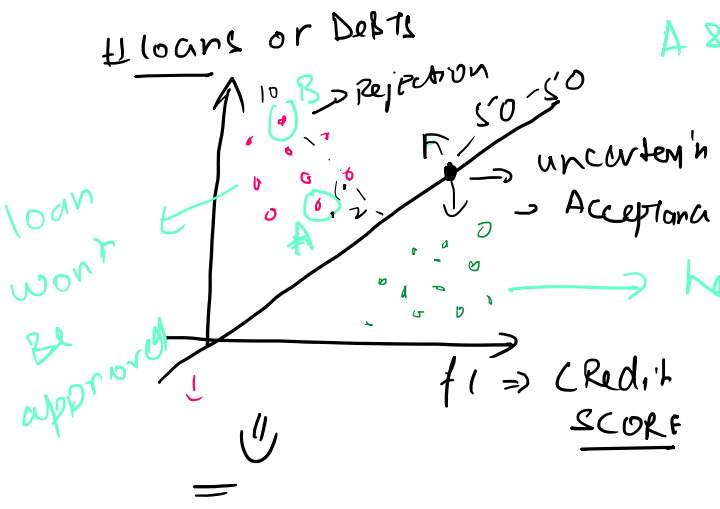
$w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

(*) $[w_1 \; w_2]^T$

Me → column vector

(*)

$\times [w_1 \; w_2]^T$



A & B ⇒ which customer has higher prob. of loan Rejection

⇒ here approval will happen

⇒ (B) ⇒ (?)

How do you know when does your model confident about prediction? line

* The points that are very far away fr will have higher prob. of that pred

& points closer to line will lesser probability

(*)

⇒ This is how we are going to use DISTANCES

Recap ⇒

Vectors ⇒ $\vec{x}, \vec{y}, \in R^d$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \quad \text{or} \quad \vec{x} = [x_1 \; x_2 \; \cdots \; x_d]$$

default → we use column vect

Norm ⇒ length / Magnitude

$$\|\vec{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots x_d^2}$$

↳ L2 Norm

Manhattan → L1 Norm

DOT product $\Rightarrow \vec{x} \cdot \vec{y} = x^T \cdot y = \|x\| \|y$

$$= \sum_{i=1}^{n} x_i \cdot y_i$$

Angle $(\vec{v}_1, \vec{v}_2) \Rightarrow \cos\theta = \dfrac{v_1^T v_2}{\|v_1\| \|v_2\|}$

unit vector $\Rightarrow \hat{a} = \dfrac{\vec{a}}{\|a\|}$

↳ to only tell about di

PROJECTION $\Rightarrow \|p\| = x^T \hat{y} = \dfrac{\|x\| \cos}{\text{scalar val}}$

vector Projection $\Rightarrow p = (x^T \cdot \hat{y})$

y ||
cos θ

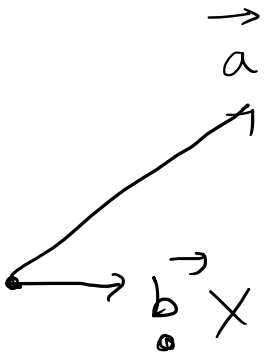
ction

s θ

ue

$\hat{y}$

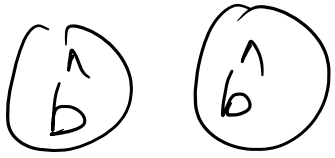
$\vec{a}$ & $\vec{b}$



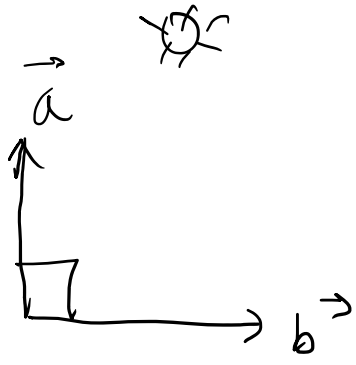
$\Rightarrow$ $\vec{a} \perp \vec{b} \Rightarrow$

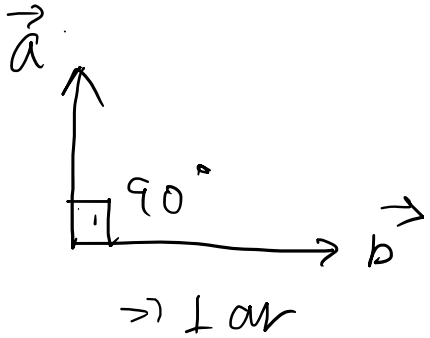
is Projection poss

$\Rightarrow$ we are not concerned about magn of



Question: If the projection of vector ā onto vector b̄ equals zero and ‖ā‖ ≠ 0, what can conclude about the two vectors?





$x_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ $x_2 = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$

Norm $\Rightarrow$ L2 $\Rightarrow$ L2 ✓





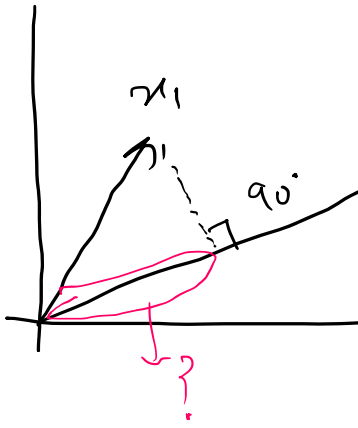
we

$$x_1 = \sqrt{2^2+3^2} = \sqrt{13}$$

$$x_2 = \sqrt{6^2+7^2} = \sqrt{36+49} = \sqrt{85}$$

$$\text{Proj.}_{x_2}(x_1) = \left(x_1 \cdot \hat{x}_2\right)$$

$$[2,3] \cdot \begin{bmatrix} \frac{6}{\sqrt{85}} \\ \frac{7}{\sqrt{85}} \end{bmatrix} = \frac{12}{\sqrt{85}} + \frac{21}{\sqrt{85}} = \frac{33}{\sqrt{85}}$$



Length of Projection

Shifting of Lines ⇒

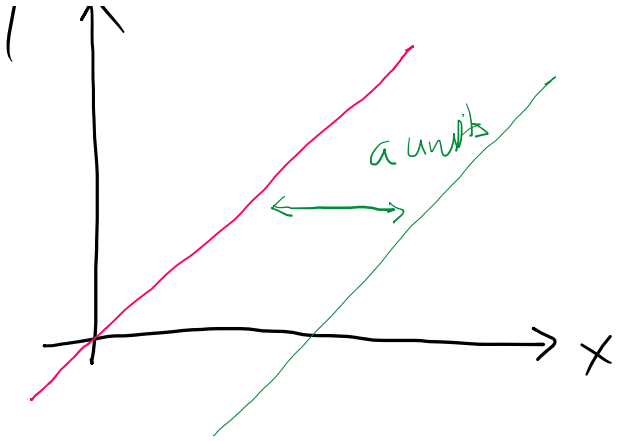
$$w_1 x + w_2 y + w_0 = 0$$

w_0 ↓ b

↓ modifications ?

Y ↑

Modifications :



$w_1 x_1 + w_2 x_2 + w_0 = 0 \Rightarrow$ eqn of Line

$(x_1 + a)$

$\Rightarrow w_1(x_1 + a) + w_2 x_2 + w_0 = 0$ → 0 (origin)

$\Rightarrow w_1 x_1 + w_1 a + w_2 x_2 + w_0 = 0$

$\Rightarrow w_1 x_1 + w_2 x_2 + (w_0 + w_1 a) = 0$

$(w_0 + w_1 a)$: constant

w_0' : new int

For changing angle of lines $\Rightarrow$ we will have to change w

cha
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ging

ont/

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$w_1 x_1 + w_2 x_2 + w_0'$ Shifted

Shifting 2D Lines

$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

How to shift a line by "a" units to the right?

$$w_1 (x_1 - a) + w_2 x_2 + w_0 = 0$$

How to shift a line by "a" units to the left?

$$w_1 (x_1 + a) + w_2 x_2 + w_0 = 0$$

How to shift a line by "a" units up?

$$w_1 x_1 + w_2 (x_2 - a) + w_0 = 0$$

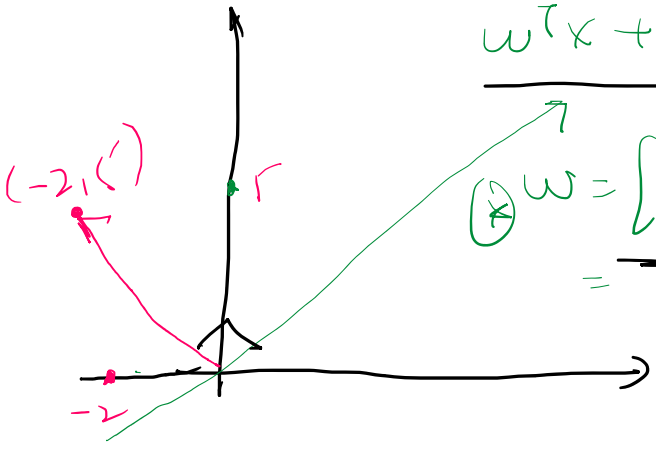
How to shift a line by "a" units down?

$$w_1 x_1 + w_2 (x_2 + a) + w_0 = 0$$

⇒ every line has a seperate $\vec{w}$

=

⇒ weight vector ORTHOGONALITY =



$w^T x + w_0$

$w = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$

=

eqn ⇒ $-2x_1 + 5x_2 + w_0$

ine

$= 0$

$\Rightarrow 2\dim V$

$= 0$

eqn ⇒ $-2x_1 + y + z$ ...

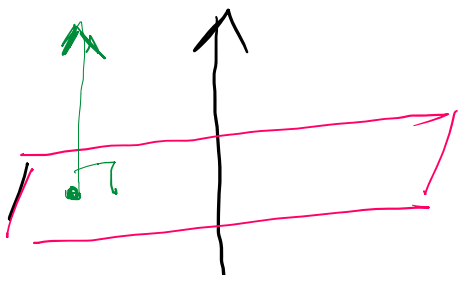
↓

(*) [if pass through or
$w_0 = 0$
otherwise $w_0 = \leq$
co

⇒ extend this concept to

$\vec{w}$ of any line
be

$$w^T x + w_0 = 0 \Rightarrow w = \begin{bmatrix} w_1 \\ w_2 \\ w_2 \end{bmatrix}$$



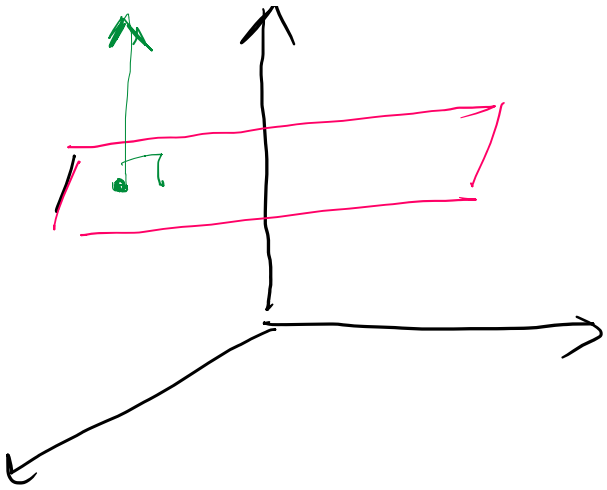
rigin

me
lue

plane as well

will always
tor to the
line

$$\begin{bmatrix} \\ \\ \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}$$



NOT

$ax + by + c = 0$

$\Rightarrow w_1 x_1 + w_1 x_2 + w_0 = 0$

Suppose $w_0 = 0$ → lin

$\Rightarrow w_1 x_1 + w_2 x_2 = 0$

$\hookrightarrow \overline{w^T x = 0}$

$E \rightarrow$ if $w_3 = 0$

$\rightarrow$ line passes through origin

is passing through origin

)

angle b/w w & $x = 90^\circ$

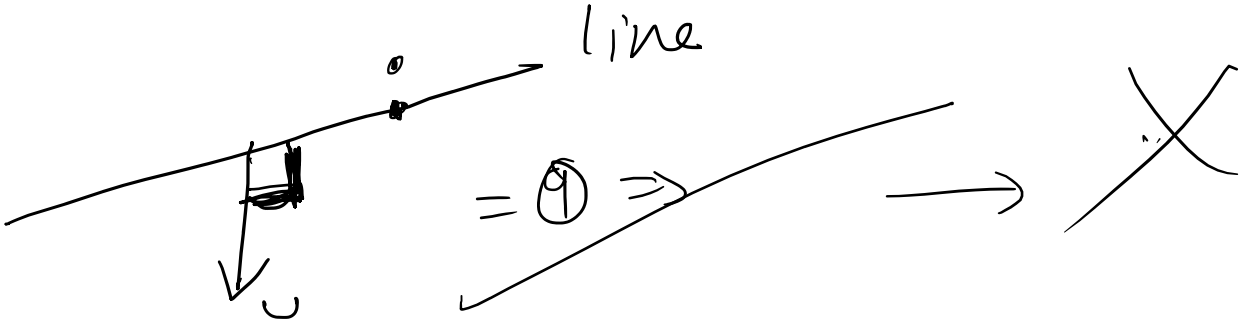
w → weights

not given a line ⇒ yo

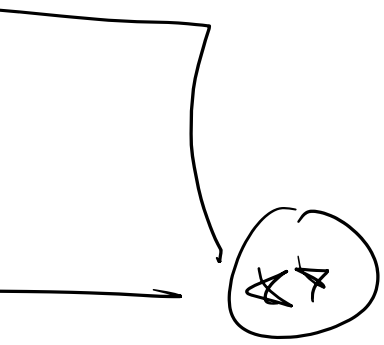
x_1, x_2 → features

features → weights → Line

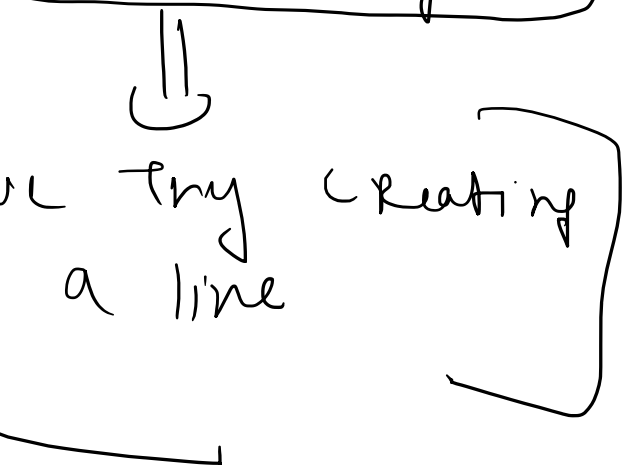
* intercept



→ line is determined by



u are given weights

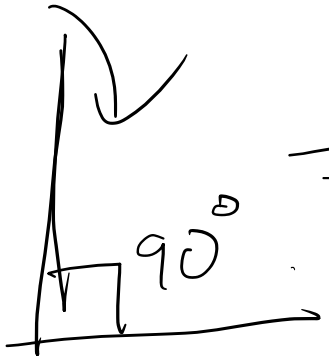


u try creating a line

weight vector

→ line is determined by

Break ⇒ 5 mins



$\Rightarrow \cos 90^\circ \Rightarrow 0$



$\begin{bmatrix} 1 \\ 2 \end{bmatrix}_{2\times 1}$ $\begin{bmatrix} \quad \end{bmatrix}_{2\times 1}$ =

T

T

$\begin{bmatrix} 1 & 2 \end{bmatrix}_{1\times 2}$ $\begin{bmatrix} \quad \end{bmatrix}$

WEIGHT VECTOR

2 → 10:28

$\cos\theta$

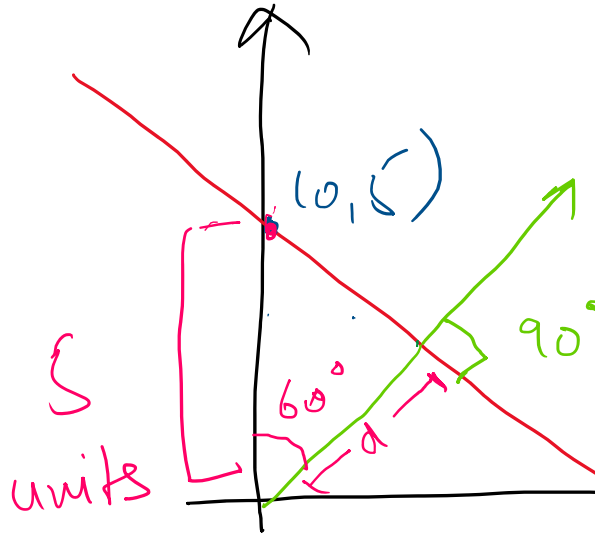
$=$

$2x_1$

→ weight vector → (Re

⇒ distance of line FROM

d = ? ⇒ dist^{n} of line from origin



$\cos 60^\circ = \frac{d}{5}$

$\Rightarrow d = 5 \times \cos 60^\circ$

$d = \frac{1}{2} \times 5 = 2.5$

ating a line

origin ⇒

$w^T x + w_0 = 0$

Line eq^n

$d = \frac{1}{2} \times 5 = 2.5$

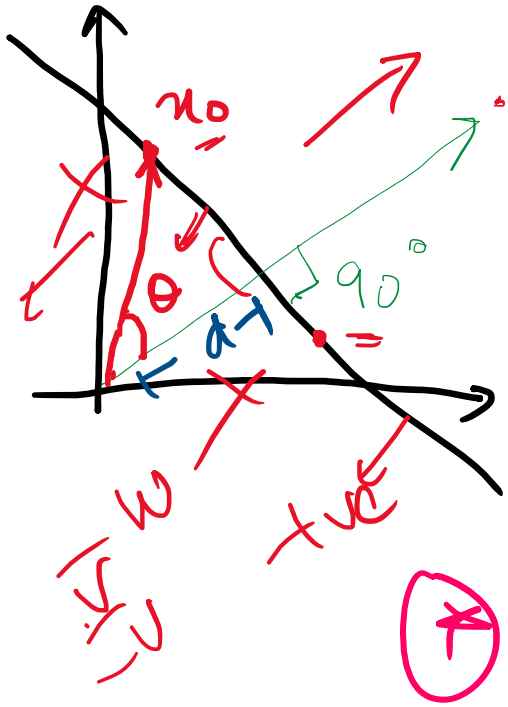
SHORTEST

→ Easy ⇒ Because I have

→ No point

Suppose

DP.



$w^T x + w_0 = 0$ ⑦ ⟵ line eqn

w

distance

given the point here

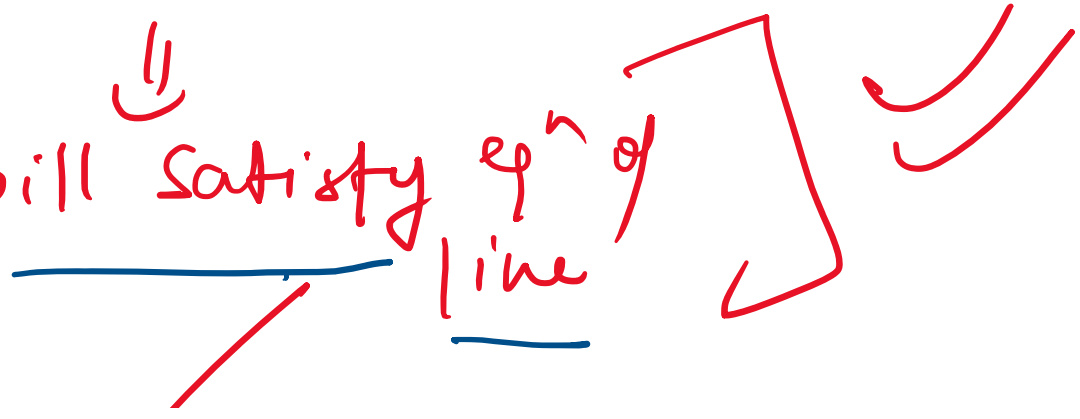
given =



there is a

(no) which lies on the line

will satisfy eqn of line



$$\Rightarrow \boxed{w^T x_0 + w_0 = 0}$$

$\cos\theta$

$$\boxed{d = \|x_0\| \cos\theta}$$

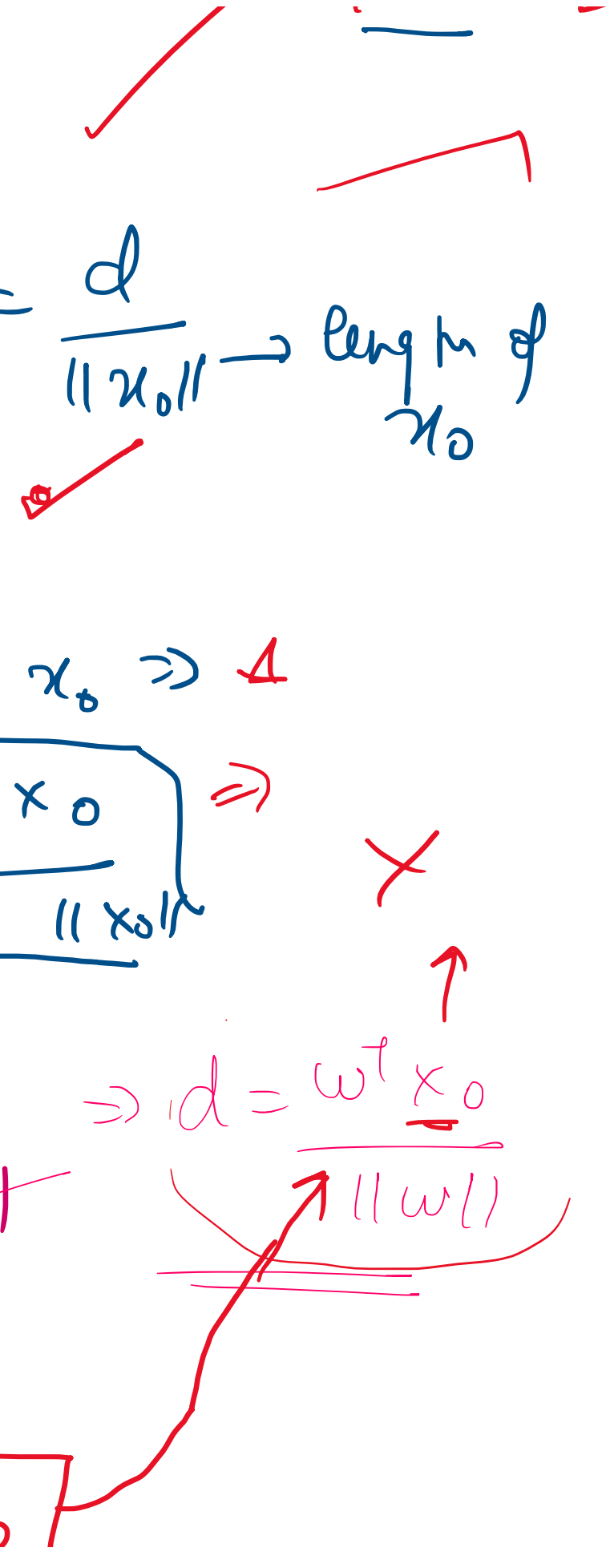
angle b/w w &

$$\cos\theta = \frac{w^T \cdot}{\|w\|}$$

$$d = \frac{\|x_0\|}{} \quad \frac{w^T \cdot x_0}{\|w\| \|x_0\|}$$

$$w^T x_0 + w_0 = 0$$

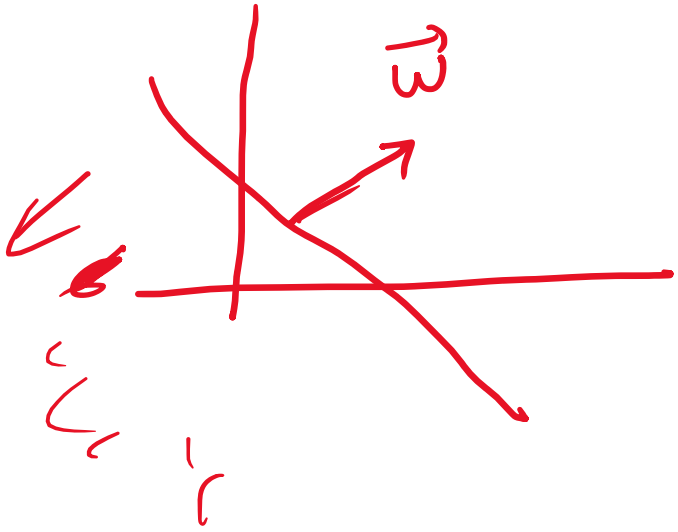
$$\Rightarrow \boxed{w^T x_0 = -w_0}$$


d
||x0|| → length of x0
x0 ⇒ Δ
x0
||x0||
⇒
X
⇒ d = ω^T x0
||ω||

$\Rightarrow \boxed{w^T x_0 = -w_0}$

Line eqⁿ

$$d = \frac{-w_0}{\|w\|}$$



$\Rightarrow$ DISTANCE of DATA

$=$

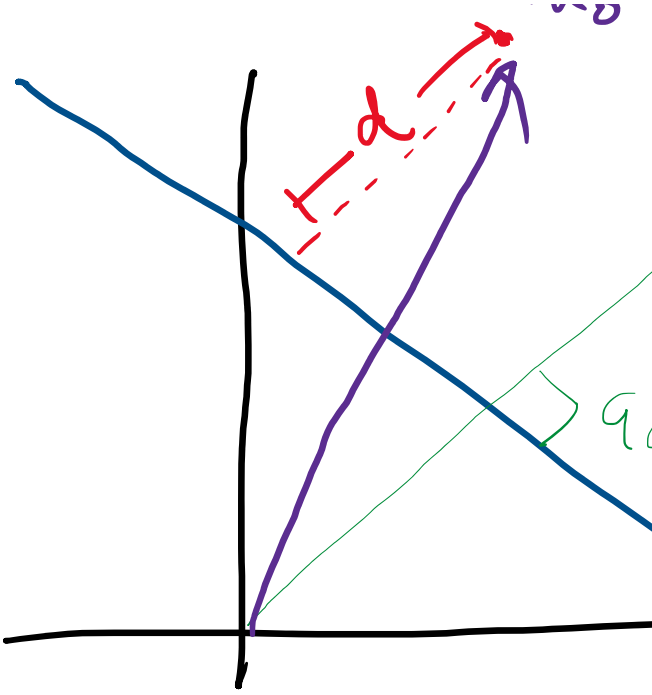


istance B/w

origin & Line] **

point from the line

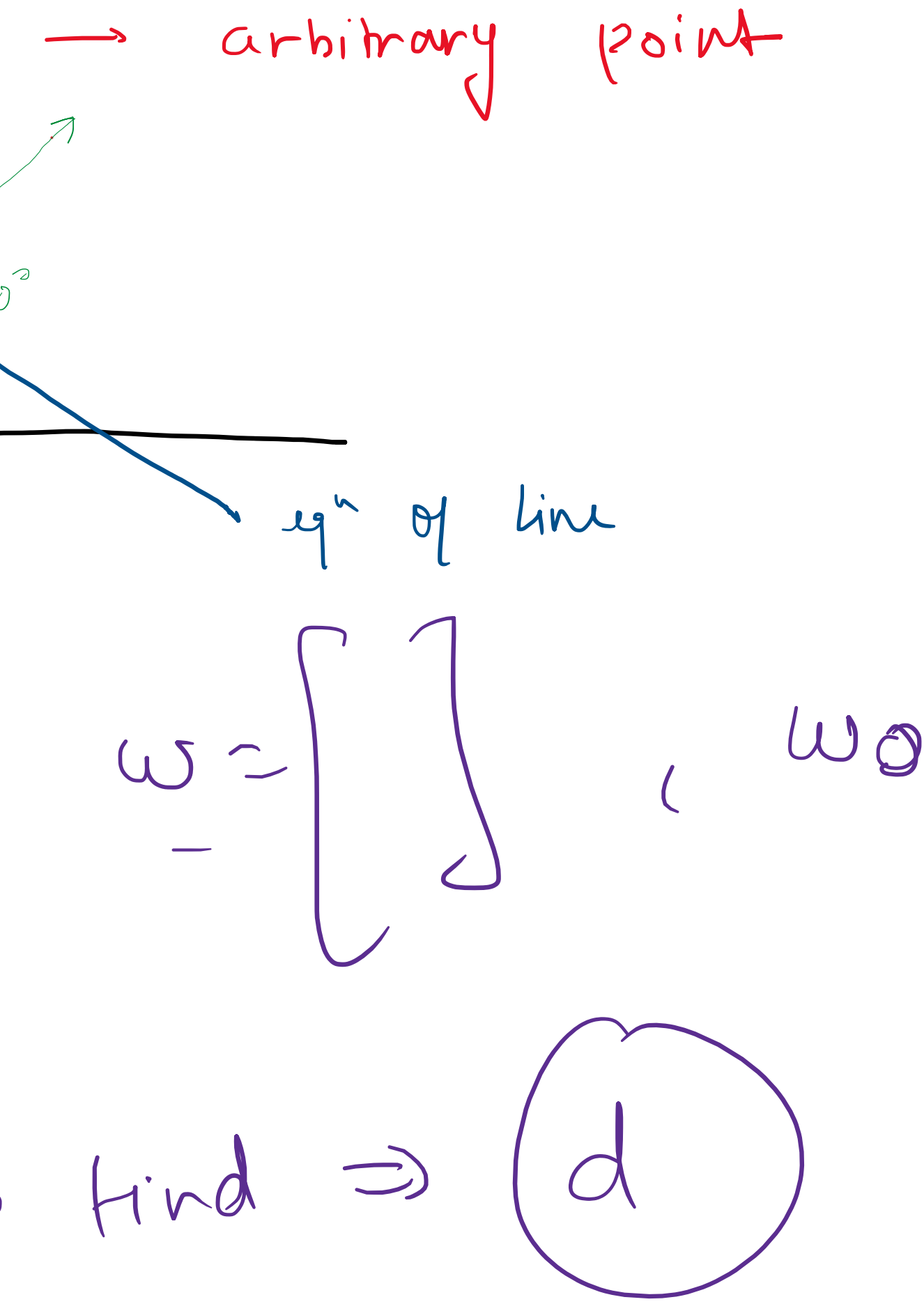
→ arbitrary point

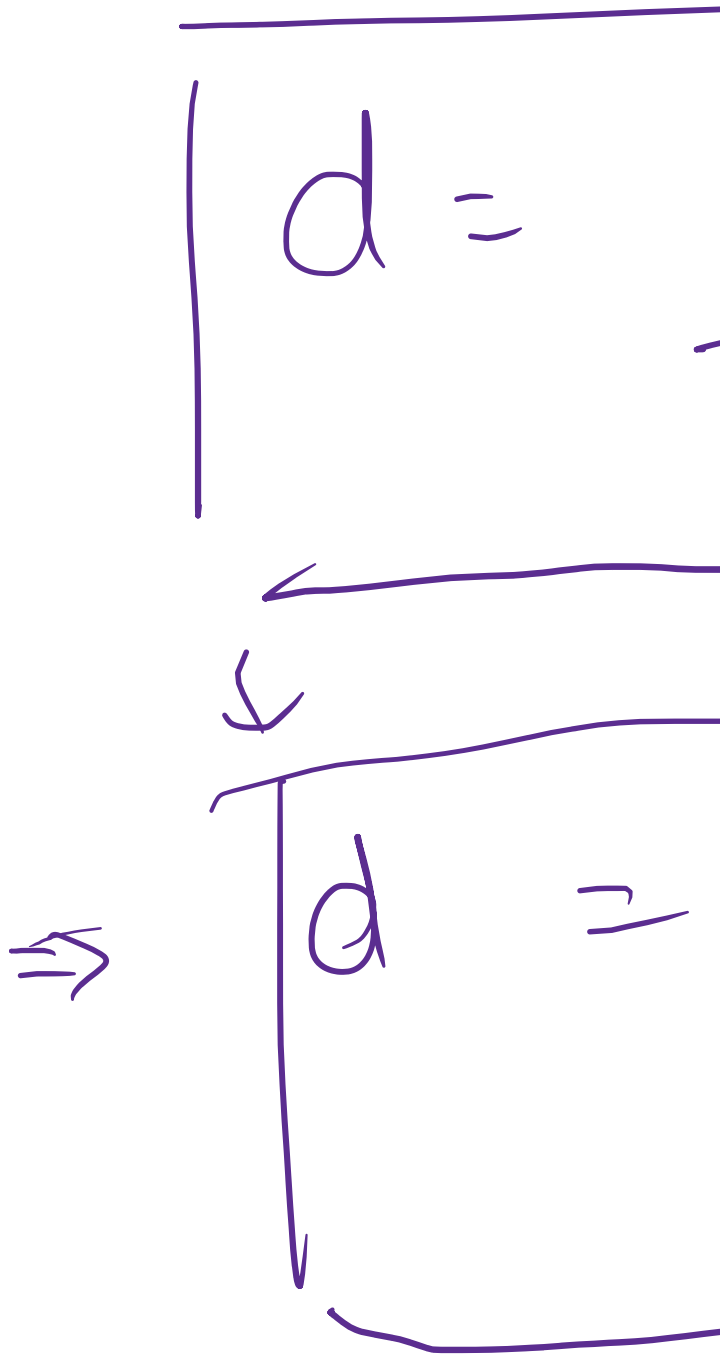


given →

$$\vec{v} = \begin{bmatrix} \\ \end{bmatrix}$$

To


arbitrary point
eqn of line
w = [ ] , w0
find
d


d =
d =

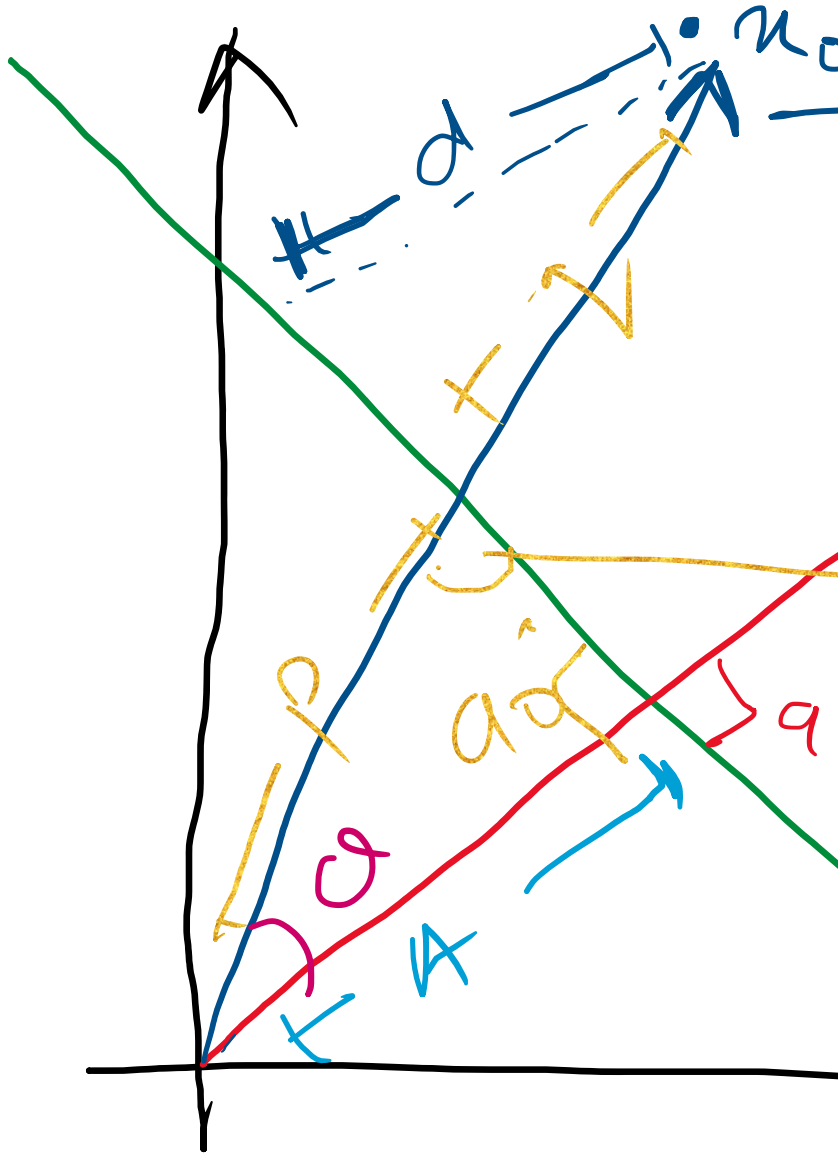
$$\frac{w^T \cdot x_0 + w_0}{\|w\|}$$

formula

$$\frac{w_1 x_{01} + w_2 x_{02} + w_0}{\sqrt{w_1^2 + w_2^2}}$$



ⓔ DERIVATION ⇒



A = distance from

of $\vec{x}$

R/w
& $\vec{x}_0$

of line

$$\|x_0\| = p + q$$

$$= \left| \right.$$

(*) $A =$ $-\omega$

$\|\omega$

$$\cos\theta = \frac{b}{H}$$

a

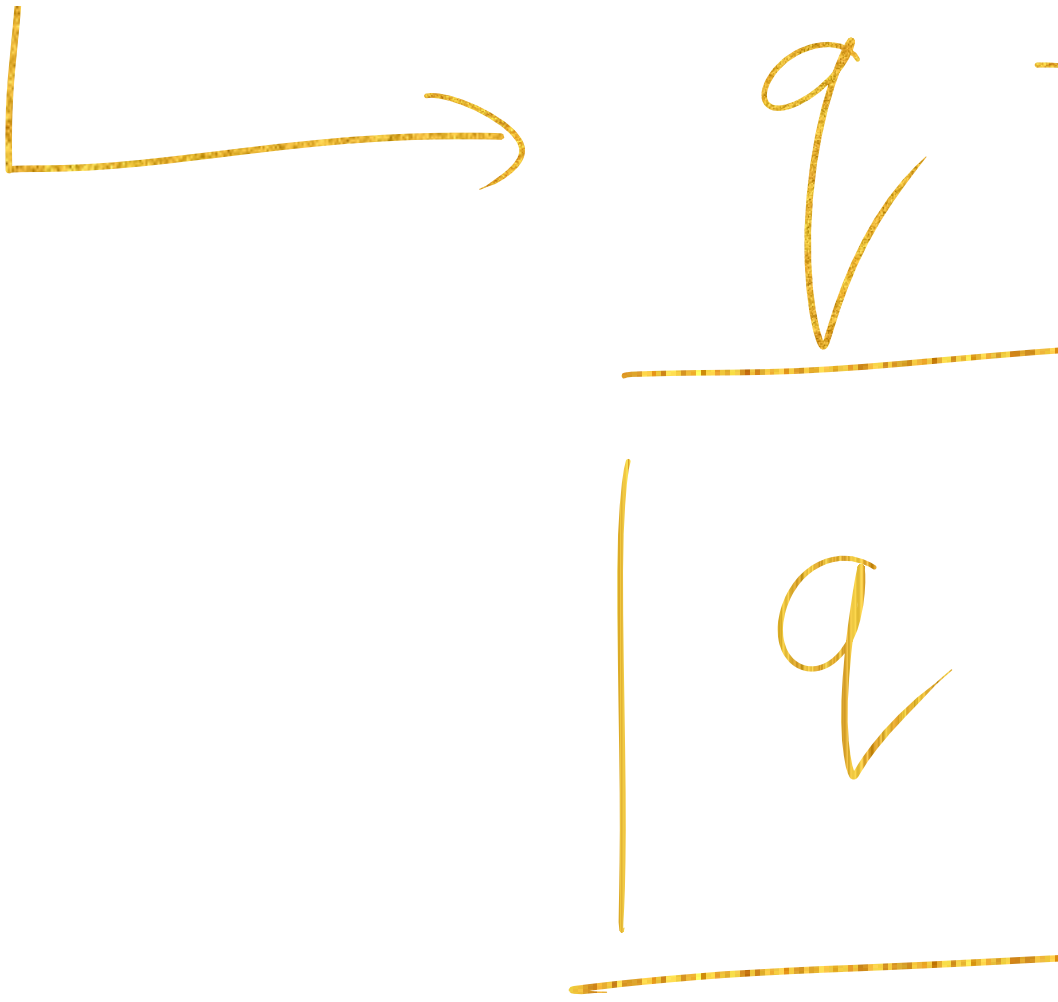
$$= \text{proj}_{-w_0}$$

$$= \frac{-w_0}{\|w\|}$$

$$\frac{}{\|\ \|}$$

$$= \frac{A}{\rho} \Rightarrow \boxed{\rho = \frac{A}{\cos\theta}}$$

$$\|x_0\| - \rho$$


upp

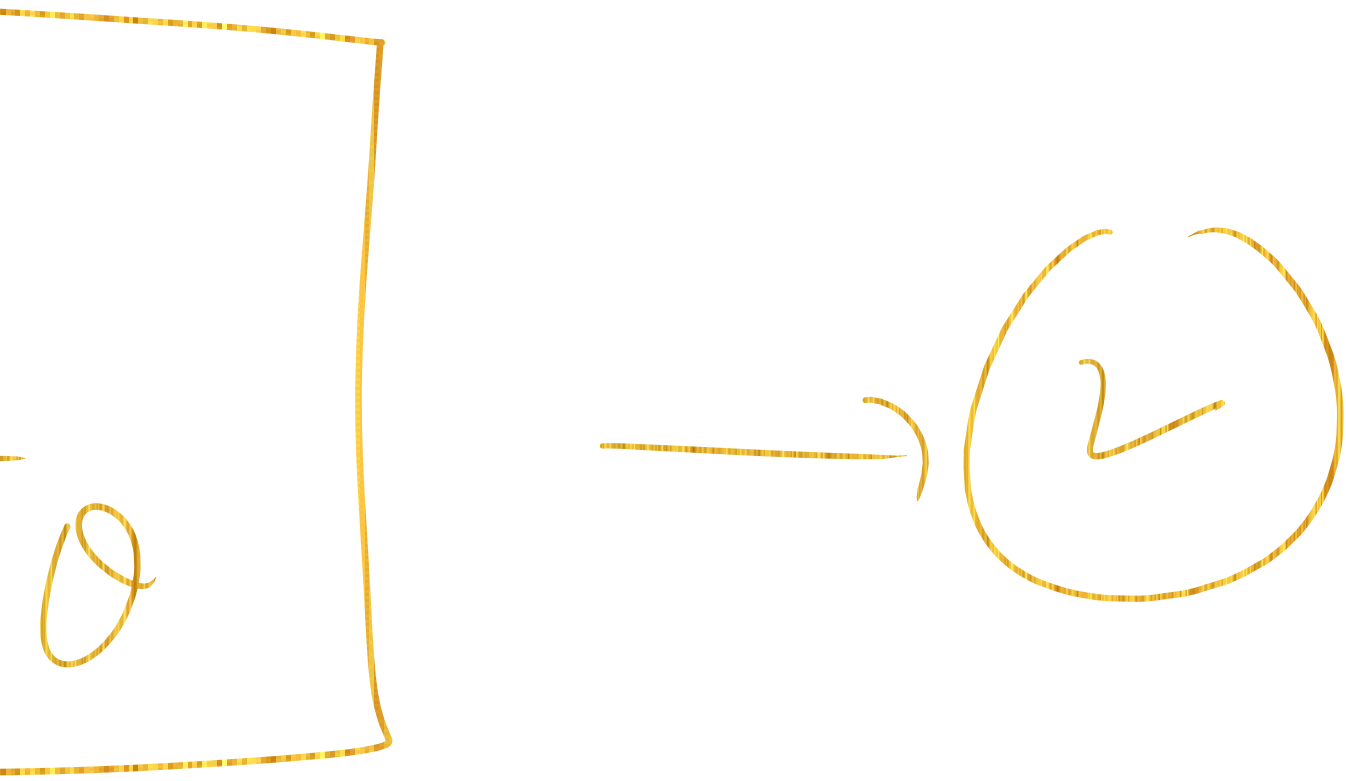
$\|x_0\| - \epsilon$

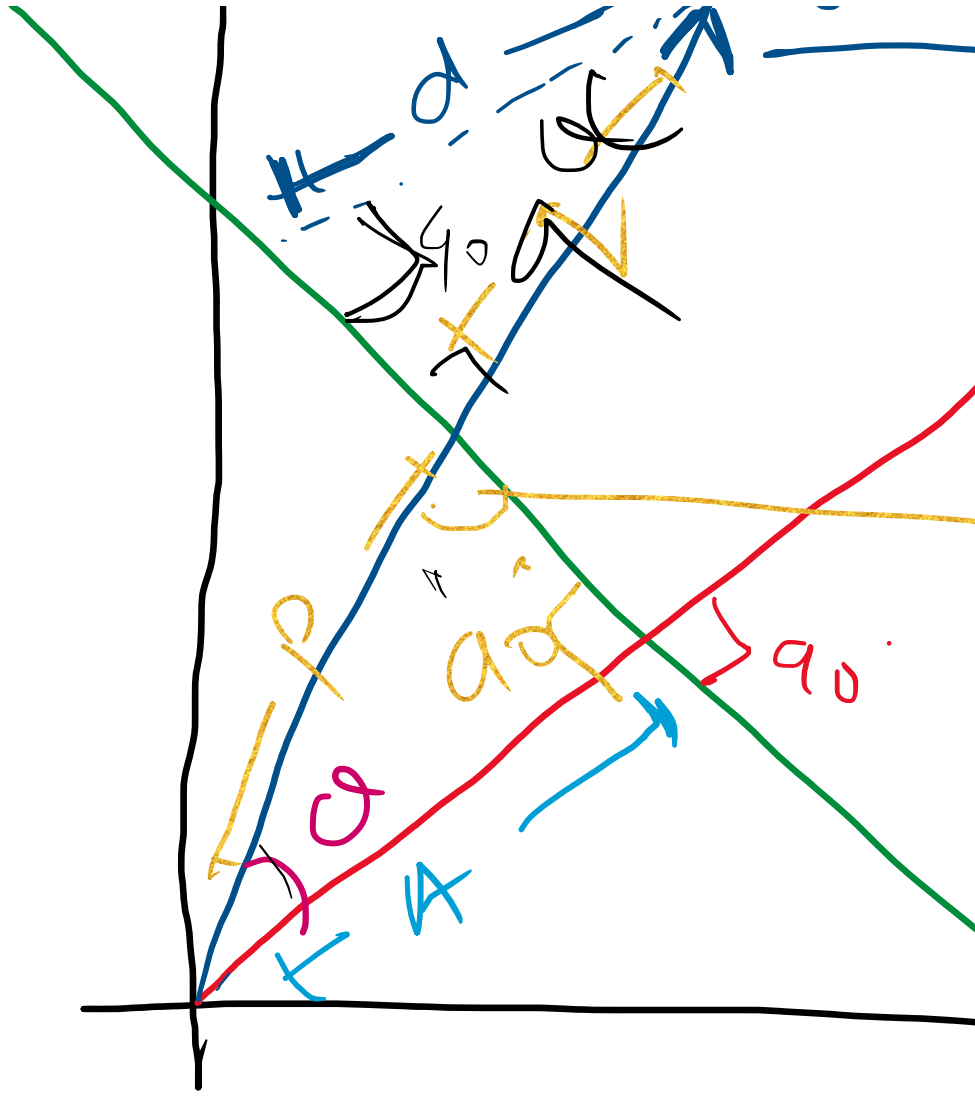
$\therefore \|x_0\| - A$

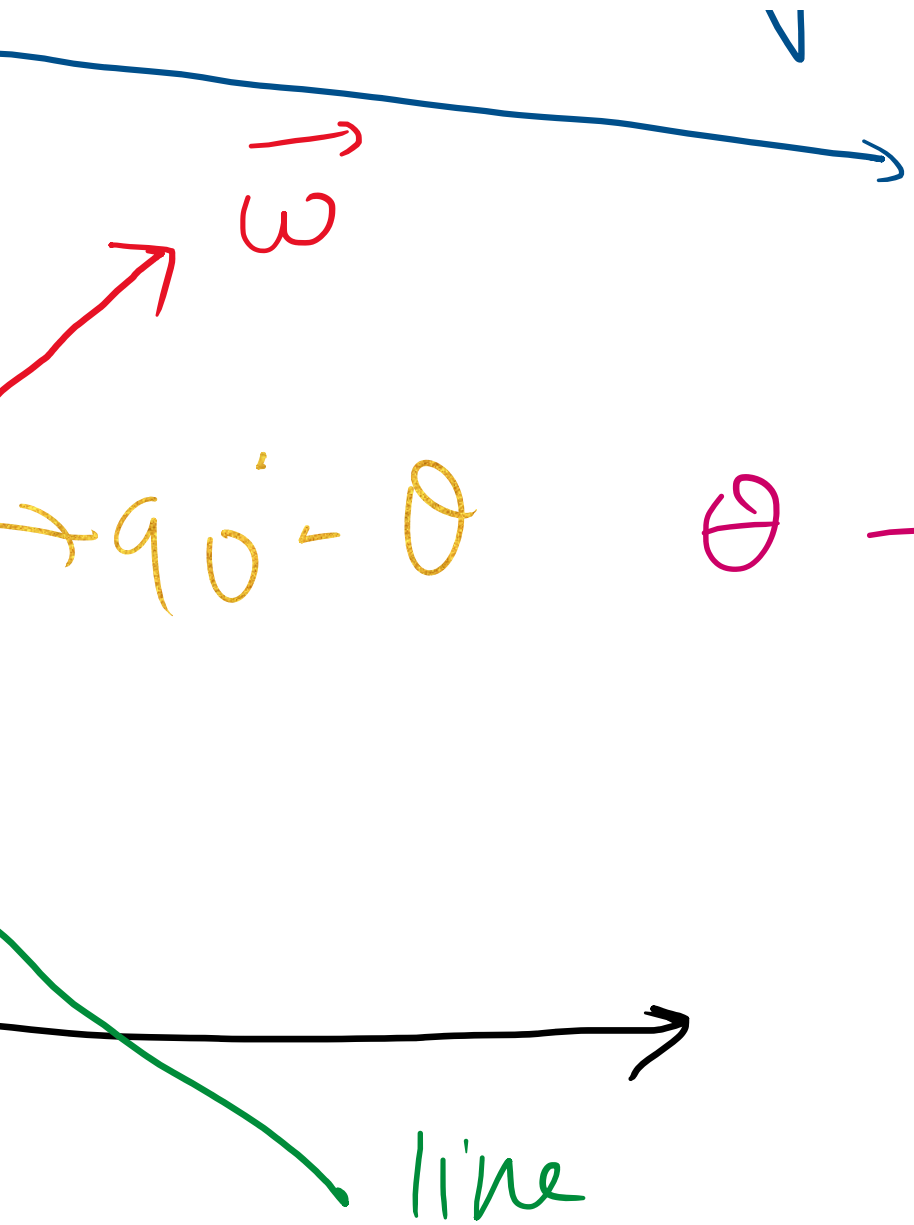
$\cos$



→ arbitrary point


0
2



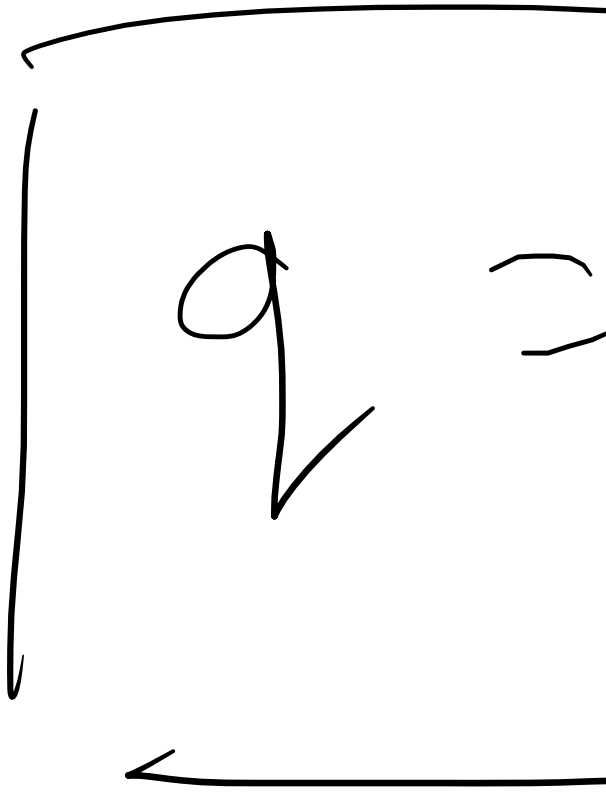


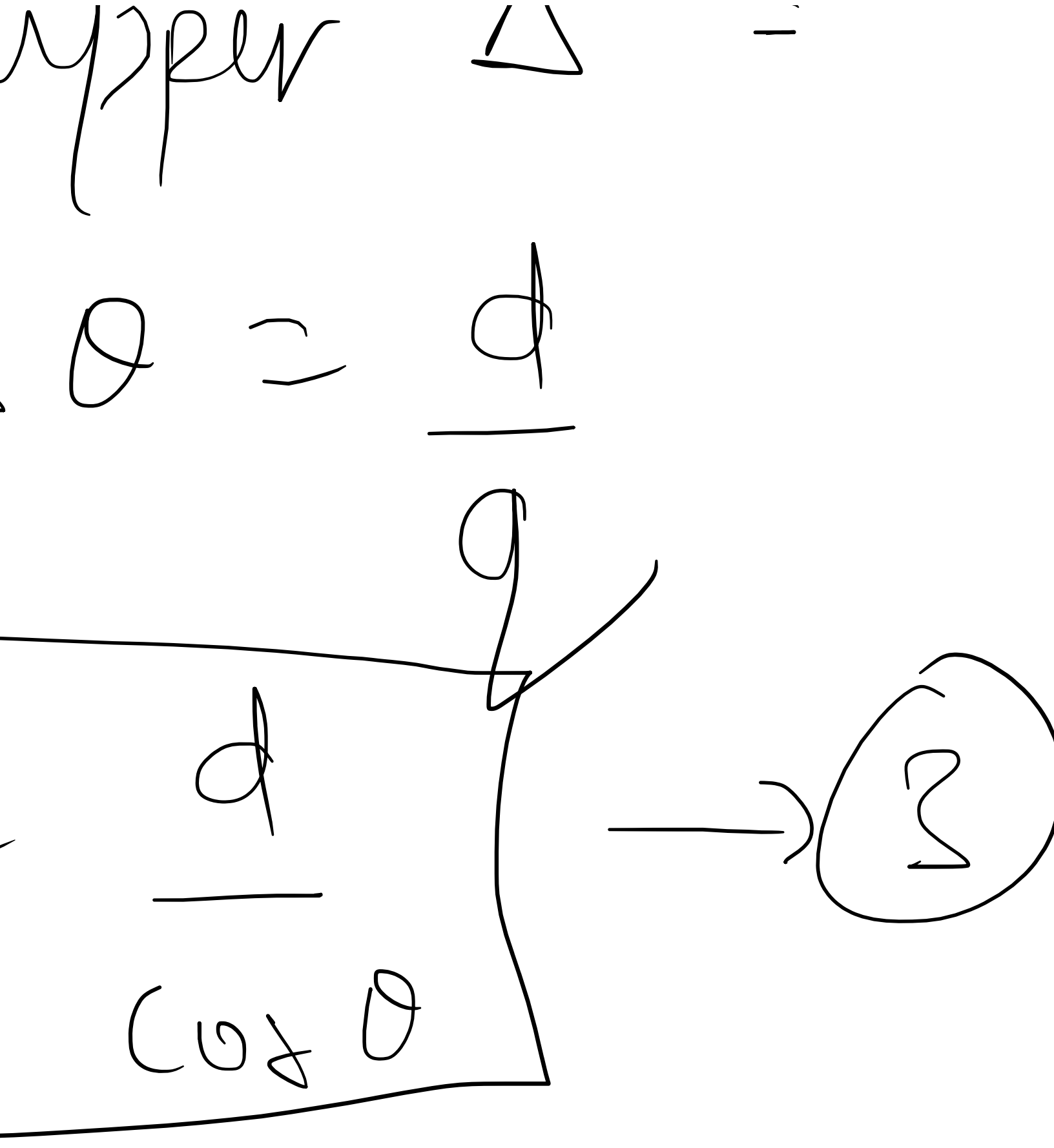
perp angles ⇒

e^{ikx}

$\vec{x}$

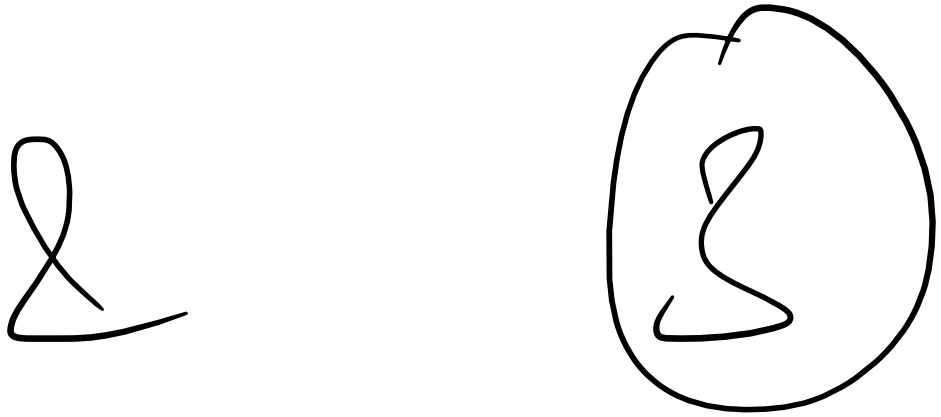
$\vec{x}_0$



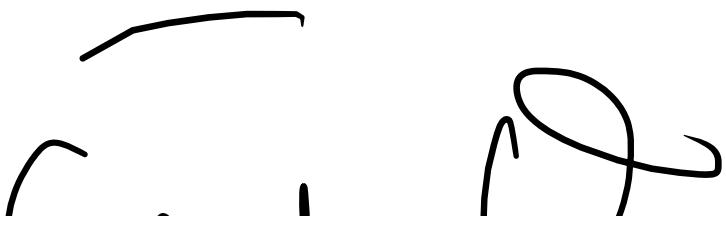

hyper
θ = d
d
cos θ

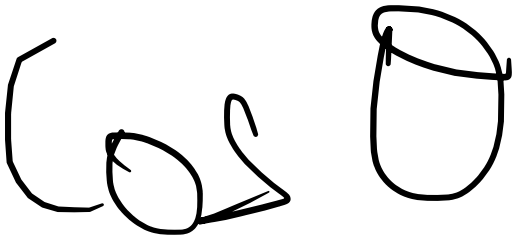
using (2)

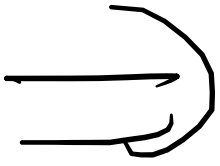
$$d =$$











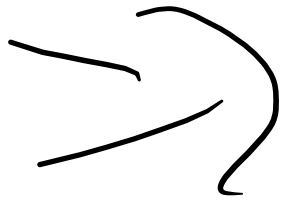
$$\int d = \ln_6$$

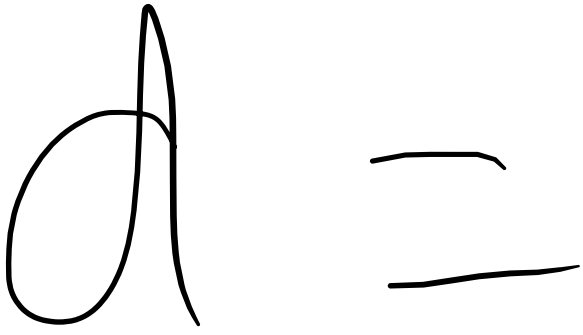
$\cos\theta$

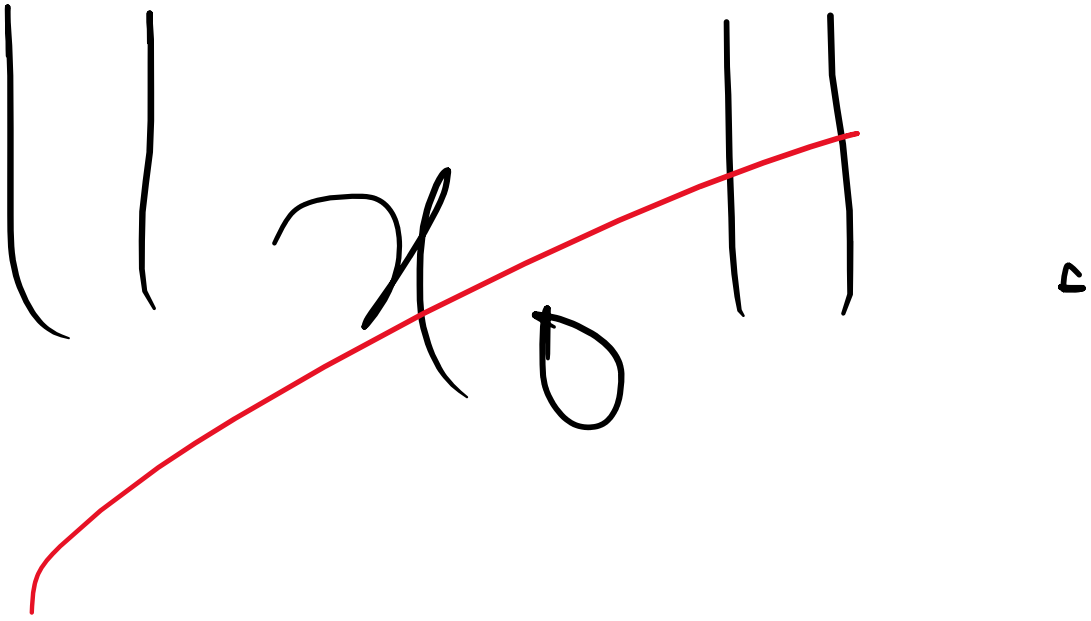
$1 - \cos\theta =$

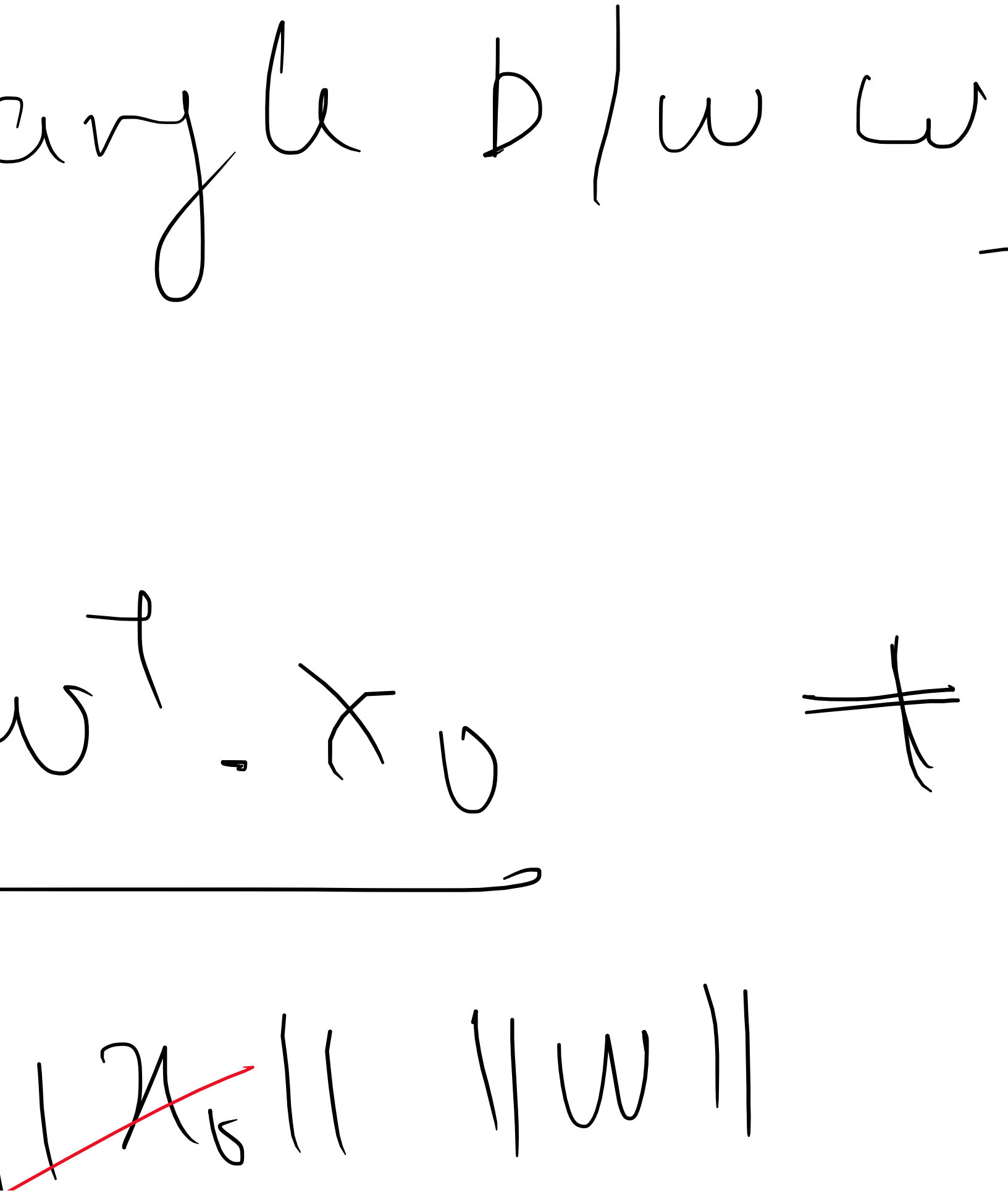
$\frac{}{\quad}$


A





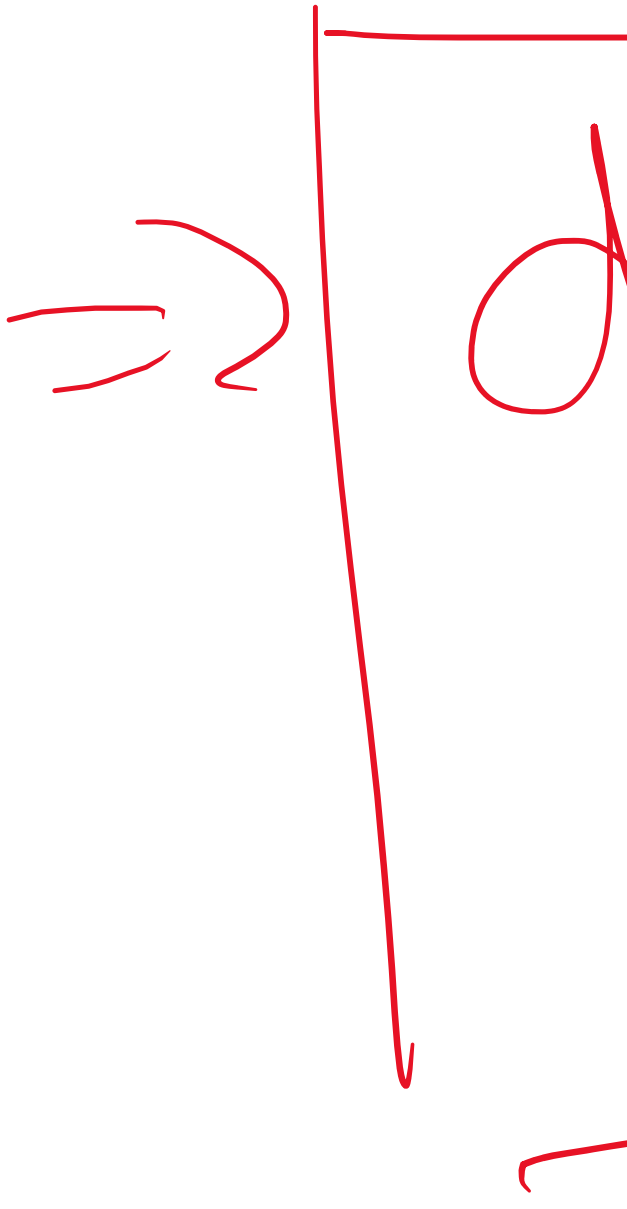


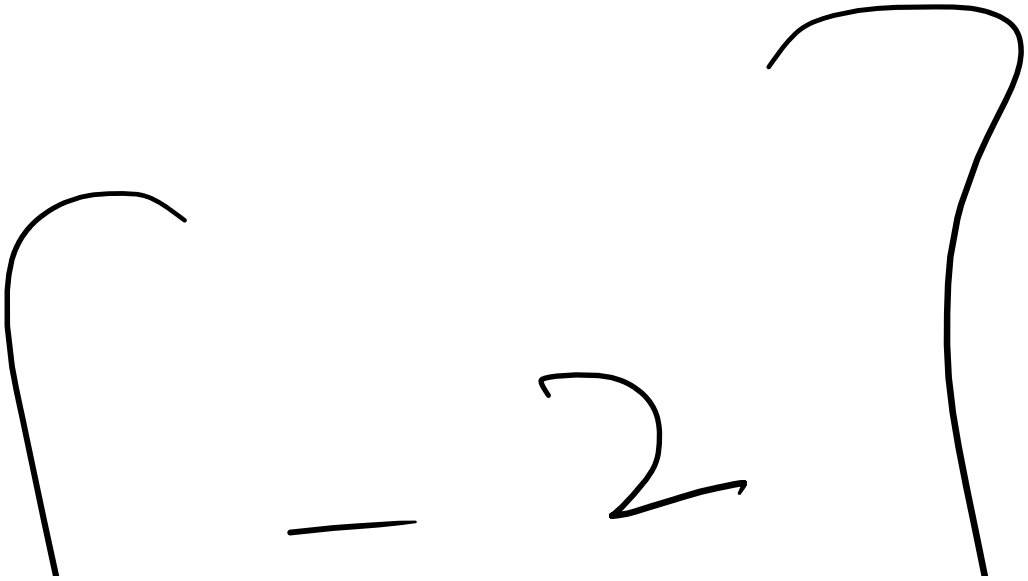


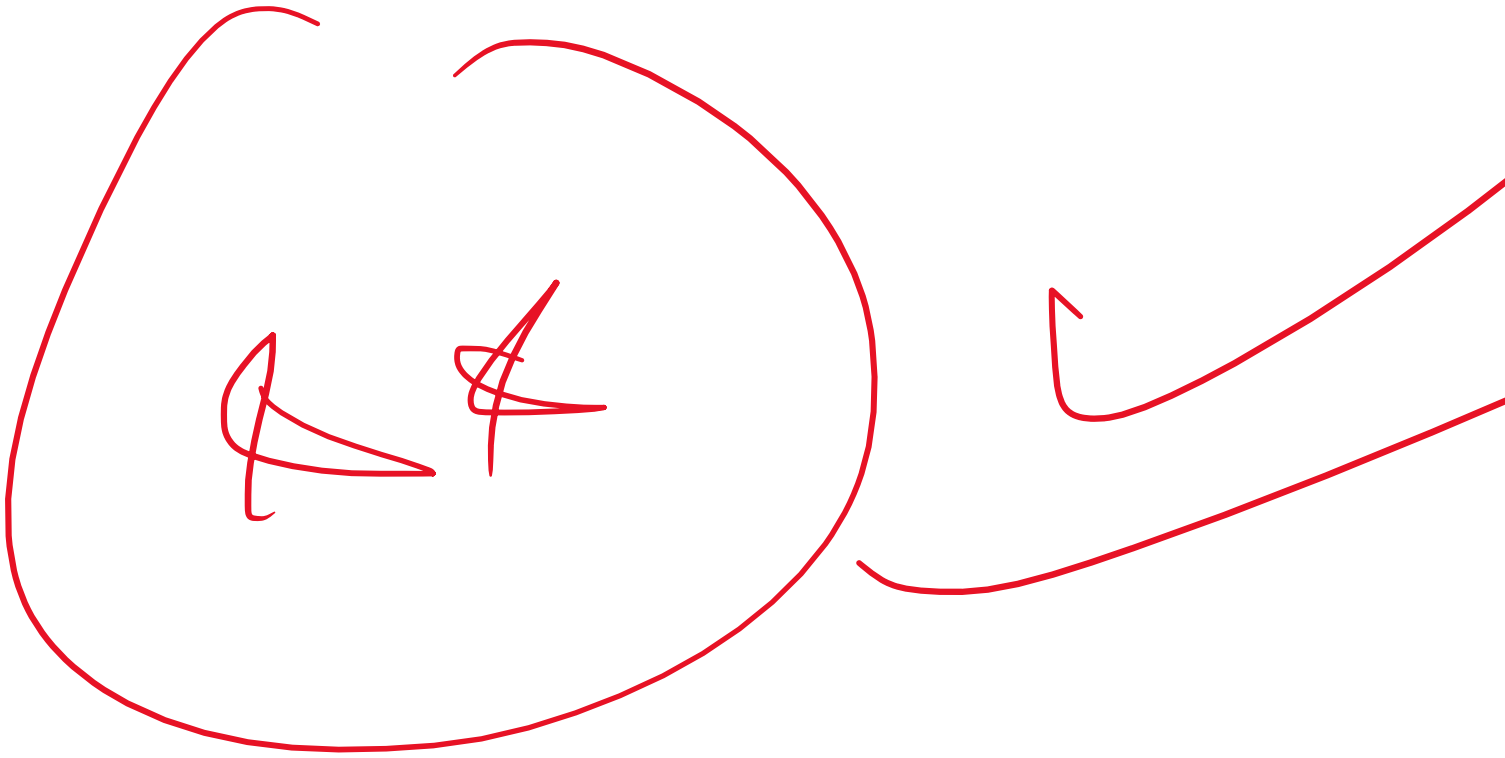
$$\text{Q} \quad H_0$$

$$\Longrightarrow$$

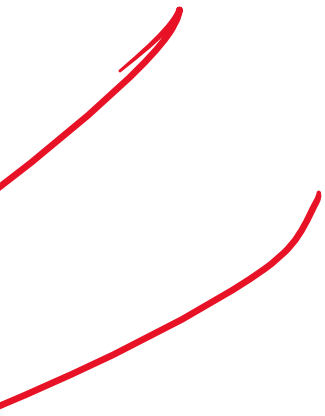
$$\frac{3 \quad 0}{11 \; 3 \; 11}$$

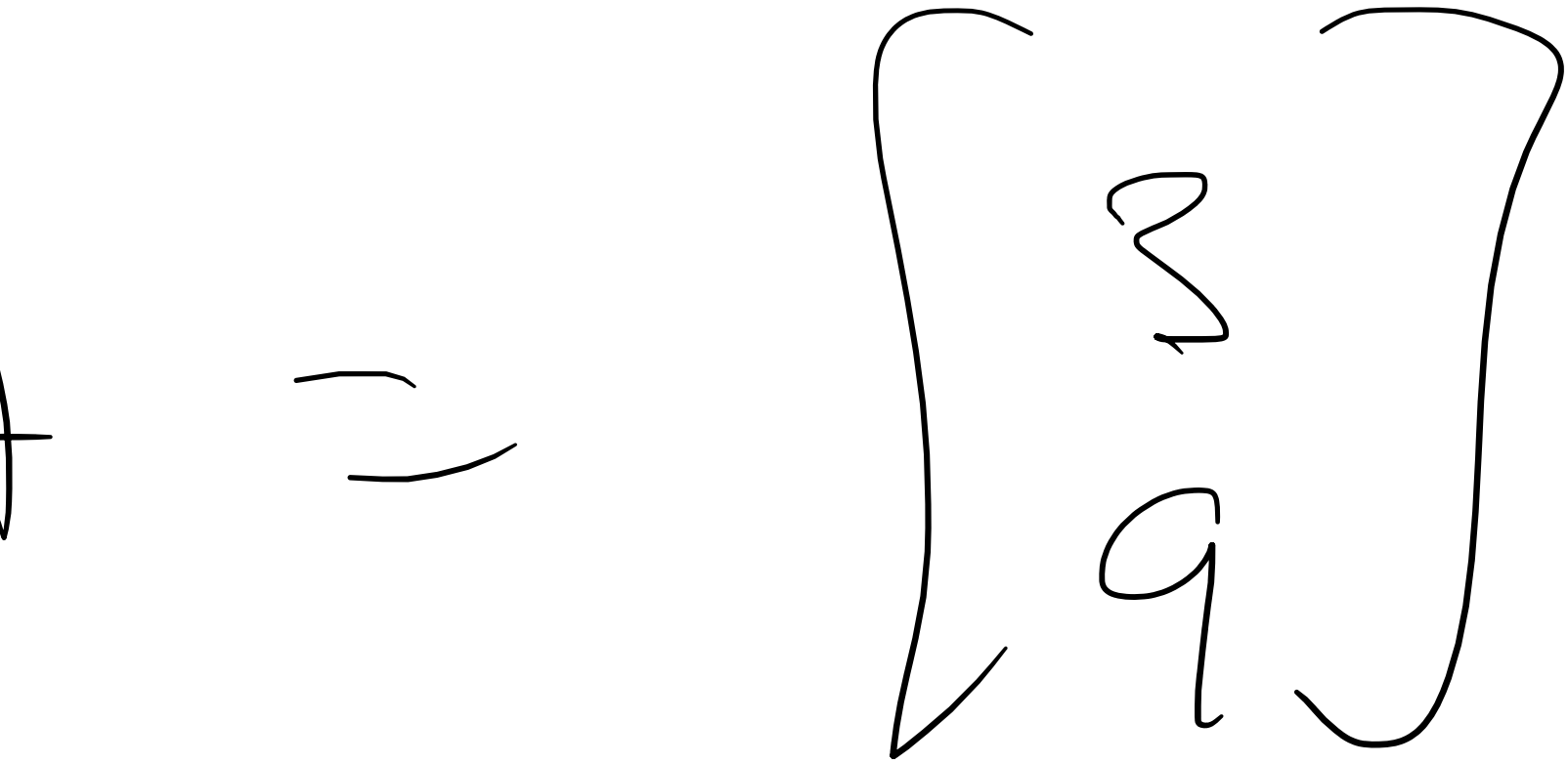










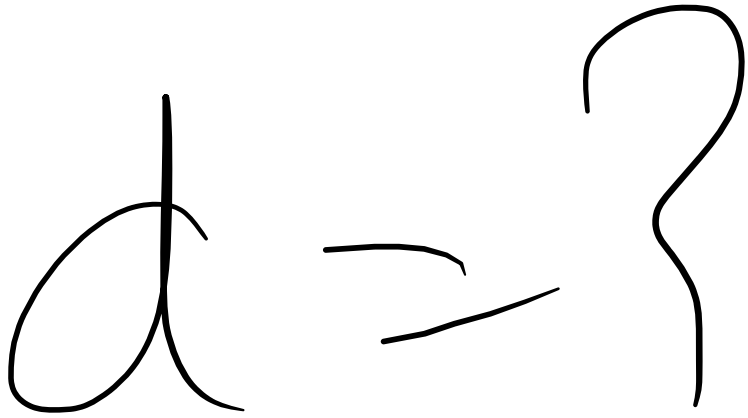


$$\omega$$

$$\omega_0$$

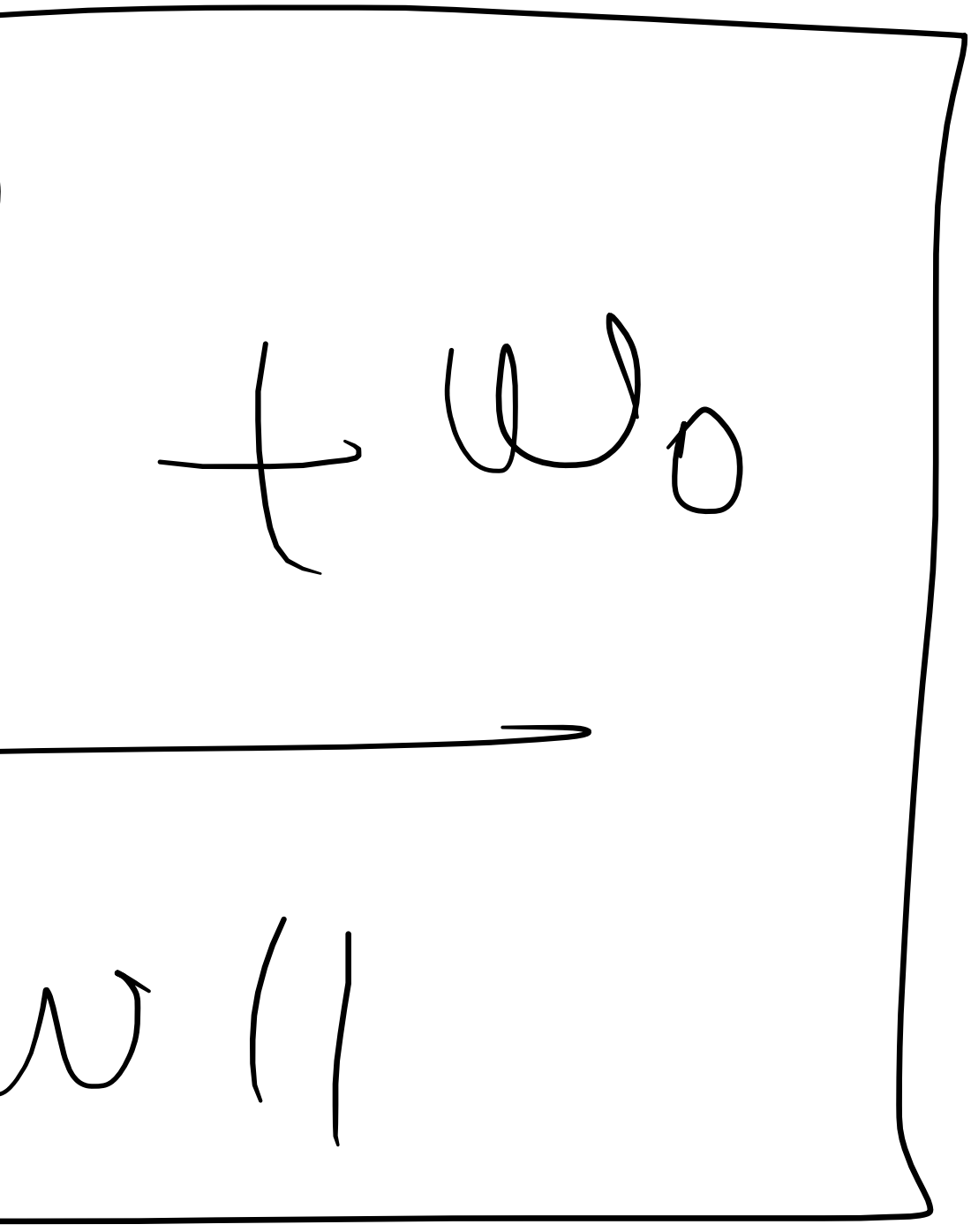
$$=$$





$$d = w^T * p$$

$$\| u$$





$$= \frac{2 - 1 - 3}{\sqrt{13}}$$

$(3, 9)$ +ve

d

